LR-schedule search for CIFAR-10 airbench

Report, 2026-10-16. Code: train_cifar.py (algorithm="auto") and research/. Every run's log is in logs/. "W-n" points to section n of interval_search_writeup.txt, the run-by-run log.

Summary

- **Algorithm:** a multi-pass knot search. Each searched LR is piecewise linear through knots every 20% of training. Pass 1 picks the knots interval by interval, scoring candidates with a lookahead to the end of training. Passes 2-3 redo the sweep using the previous pass's schedule as the lookahead and score candidates on 2 fresh seeds.
- **Batch size 2000:** matches the baseline (93.94) with no transform and under both transforms: 93.88-93.96 when searching one LR (~600 s), 93.81-93.96 when searching all four from scratch (~1800 s).
- **Batch size 500 and 125:** not yet. 0.1-0.25 pp short of the new baselines (94.01 / 93.88). Those runs used the older, weaker baselines; reruns on the new ones are in progress.
- **Main findings:** short lookaheads pick LRs that are too small; the final interval must anneal to 0; single-seed searches suffer a winner's curse; a fixed lookahead shape breaks under transforms, and the previous pass's schedule fixes it.

1. Question

Find one LR-schedule search algorithm that (a) matches the baseline's held-out accuracy, (b) uses as little search time as possible, (c) still works when an unknown transformation is applied on top of the schedule, and (d) still works at batch sizes 500 and 125 (development was at 2000).

Scenario	What is searched	Starting point
A: good baseline	One LR schedule (conv). Also tested: head only; conv + head.	The baseline's own LR; other groups stay on the baseline
B: from scratch	All four trained LRs: conv, head, norm_bias, whiten_bias	Every LR at 1.0; no knowledge of scale or shape

2. Definitions

Term	Meaning
Spec schedule	The LR schedule the search writes (piecewise linear; see section 3).
Transform	A fixed multiplier applied to the spec LR before training uses it, unknown to the search (x = fraction of training). **ramp**: x 4^(2x-1), from x0.25 to x4. **wave**: x 2^sin(2 pi x), x1 -> x2 -> x0.5 -> x1. With a transform the best spec schedule is no longer linear decay (figure 1).
Held-out TTA	Quality of a found schedule: retrained from scratch on seeds 100-107; mean test-set TTA accuracy. Search decisions also use the test set (intentional).
Search seeds	Independent runs of a search; tables give the mean held-out TTA over n of them.
Cost	Wall-clock seconds of the search on one A100, held-out retraining excluded. One full training run at batch size 2000 takes ~10 s.
Lookahead	To score a candidate LR for the current interval, train the interval, then keep training with an assumed future schedule; the TTA at the end is the score.
Do-nothing	The baseline schedule kept unchanged under the transform (no search): what a search must recover.

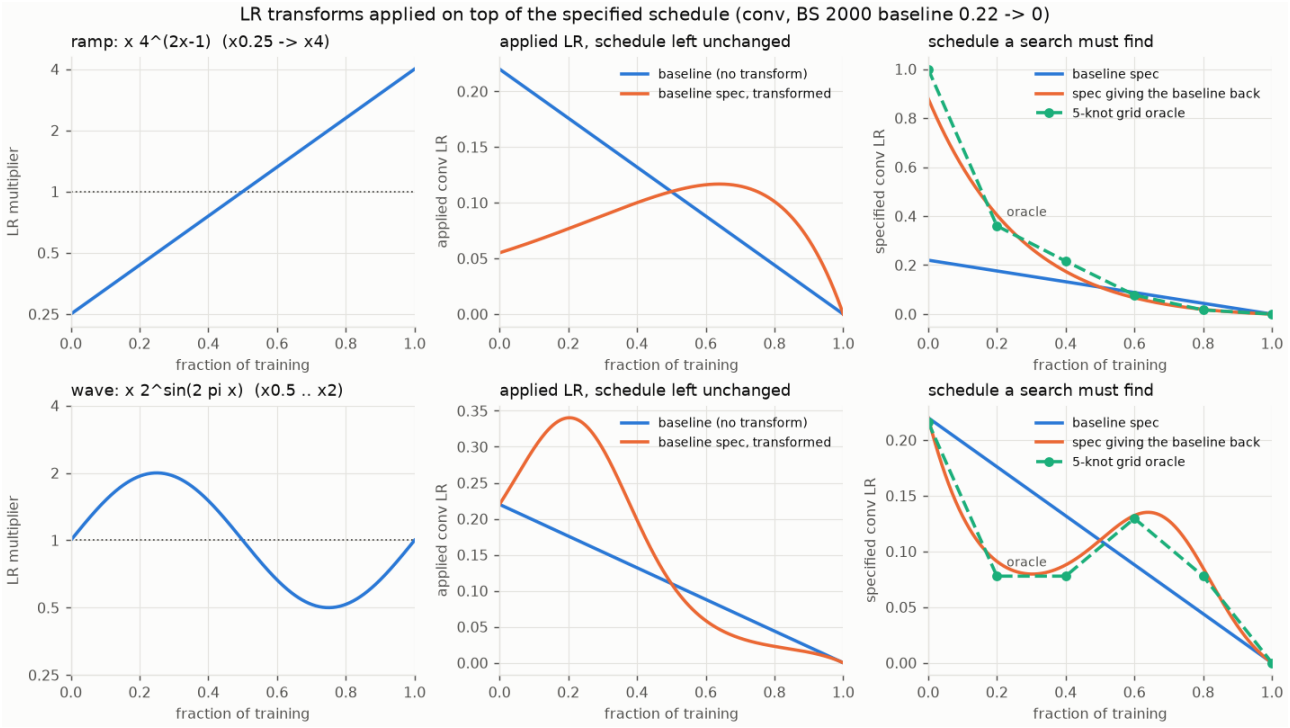


Figure 1. The two transforms (left), the applied conv LR if the baseline's linear decay is kept (middle), and the spec schedule that gives the baseline back with the grid-snapped 5-knot oracle (right). Batch size 2000 conv baseline: 0.22 decaying linearly to 0.

3. Recommended algorithm: multi-pass knot search

Configs: `auto_a_v1x3m2t_*` (scenario A) and `auto_b_v1x2m3fas_*` (scenario B) in train_cifar.py.

Representation

Each searched LR is piecewise linear through 6 knots, one at every 20% of training (5 intervals). Knot values lie on the grid 0.6^k rounded to 2 significant figures.

Pass 1: one forward sweep over the 5 intervals

1. For each searched LR, coordinate-climb the knot at the end of the current interval on the grid, starting from the previous interval's value: try one step up and one step down, keep moving while the score improves.
2. Score a candidate by training the interval from the committed state, then a lookahead to the **end of training** in which the searched LRs decay linearly from the candidate value to 0; the other groups follow their own schedules.
3. Commit the interval with the best knot and continue from the trained state.

Passes 2 and 3: the same sweep, restarted from the initial weights

- Each interval starts its climb at the previous pass's knot.
- The lookahead is the previous pass's schedule for the rest of training, rescaled by candidate / previous knot, so one decision shifts the whole remaining schedule.
- Every candidate is scored as the mean over 2 fresh training seeds.

Additions for scenario B

- First interval: before climbing, scan each LR over 7 values from x10^-3 to x10^3 around its start (with a 20% lookahead) and keep the best.
- Last interval: also try annealing every searched LR linearly to 0; keep it if it scores higher.
- Two passes instead of three; pass 2 scores on 3 seeds and does not rescale the tail.

4. Results

Held-out TTA (%), mean over n search seeds, with search cost in seconds in brackets.

4.1 Scenario A: conv searched, other groups on the baseline

Batch size	Method	No transform	Ramp	Wave
2000	Baseline	93.94		
	Do-nothing		93.42	93.72
	Recommended (3 passes), n=4	**93.96** (588)	**93.93** (611)	**93.88** (572)
	2 passes, n=4	93.98 (359)	93.78 (387)	93.91 (362)
	Pass 1 only, n=2	93.94 (132)	93.57 (150)	93.59 (128)
500	Baseline: old / new	93.83 / 94.01		
	Do-nothing (old baseline)		93.35	93.68
	Recommended, n=2	93.79 (725)	93.75 (774)	93.84 (736)
125	Baseline: old / new	93.85 / 93.88		
	Do-nothing (old baseline)		93.31	93.68
	Recommended, n=2 (ramp n=1)	93.64 (1159)	93.88 (1109)	93.72 (1164)

Other searched groups (batch size 2000, n=2)	No transform	Ramp	Do-nothing ramp
Head only	93.91	93.90	93.82
Conv + head	93.99	93.73	93.30

4.2 Scenario B: all four LRs from 1.0

Batch size	Baseline	No transform	Ramp	Wave	n
2000	93.94	93.96 (1690)	93.81 (1867)	93.91 (1814)	2
500	94.01	93.89 (2471)	93.90 (2698)	-	1
125	93.88	not run yet			

At batch size 2000, do-nothing under ramp scores 93.25 and one-pass variants 93.05-93.36.

4.3 Verdict

- **Batch size 2000: goal met** in both scenarios and all three conditions (within 0.13 pp, mostly within 0.05 pp), at ~600 s per searched LR (~60 training runs) or ~1800 s from scratch. The earlier 8-seed global search cost 3780 s for four groups and cannot change a schedule's shape.
- **Batch size 500 and 125: not met.** Against the new baselines, scenario A is 0.1-0.25 pp short at 500 and 0.0-0.25 pp short at 125.

4.4 Baseline update

After the campaign, baseline_bs500 and baseline_bs125 were replaced by the best SGD configs of the earlier seed-averaged searches (same optimizers: Muon2 conv, SGD head and biases): batch size 500 94.01 +- 0.12 (seeds 100-115; the previous baseline scored 93.83 on seeds 100-107), batch size 125 93.88 +- 0.11 (previous 93.85, a tie). All batch size 500 / 125 numbers above used the previous baselines, as the target and as the fixed groups of scenario A. Reruns on the new baselines are in progress; do-nothing on them: batch size 500 ramp 93.37, wave 93.77; batch size 125 ramp 93.48, wave 93.69.

5. Findings

F1. Short lookaheads choose LRs that are too small, whatever the cooldown (W-1, W-4)

At interval 2 (batch size 2000) the final accuracy is best for conv LR 0.13-0.22, but every 40-step lookahead (72 cooldown designs tried) scores 0.078 best. An 80-step lookahead picks 0.13 by a hair, and a lookahead to the end of training picks 0.13 (figure 2).

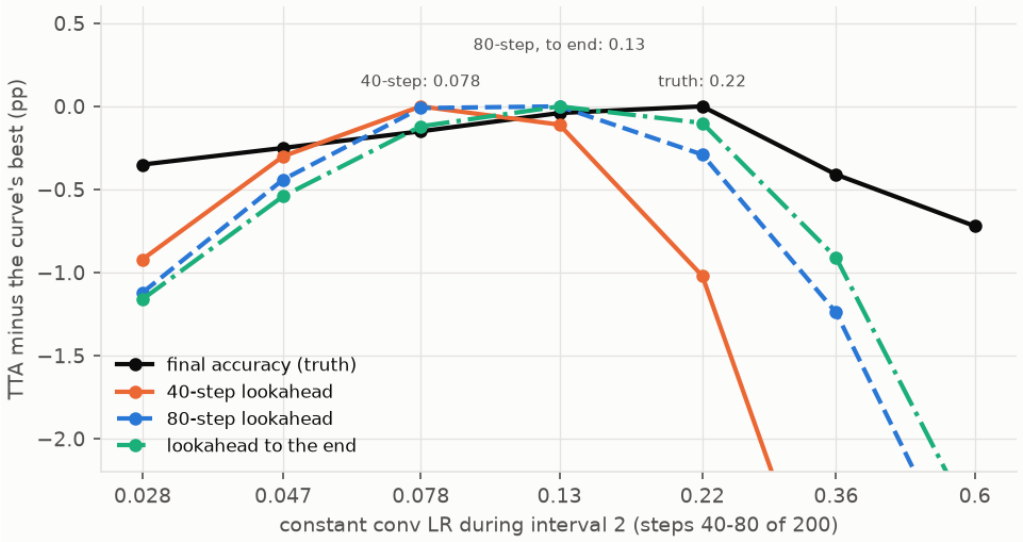


Figure 2. Each curve relative to its own best value; labels give the LR each curve would choose. 4 seeds per point.

F2. The non-searched groups' cooldown does not matter (W-1, W-5)

Continuing their schedule, decaying to 0, or any constant multiplier (even 0) changes proxy scores by at most 0.1 pp and never changes the ranking. Full searches agree: 93.73-93.78 for all three designs.

F3. The final interval must anneal to 0; constant intervals are otherwise fine (W-6)

The baseline as 5 constant steps scores 93.64; the same with the last step decaying to 0 scores 93.90; 10 constant steps score 93.92. The 5-knot grid is enough at every batch size: grid-snapped baseline knots score 93.98 / 93.84 / 93.83 at batch sizes 2000 / 500 / 125.

F4. A fixed lookahead shape breaks under transforms (W-7, W-8)

Pass 1 alone scores 93.57 under ramp and 93.59 under wave (do-nothing 93.42 / 93.72). Under ramp the linear lookahead applies too much LR late, which penalizes high early LRs: the search chose 0.13 where about 0.9 is right. Searching the lookahead's level or a geometric shape helped some transforms and hurt the untransformed case (93.74).

F5. The previous pass's schedule is a self-correcting lookahead (W-12)

Two passes score 93.94 under ramp. Late decisions have short lookaheads and are nearly unbiased, so each pass corrects the lookahead of the earlier intervals.

F6. Single-seed searches suffer a winner's curse (W-12, W-13)

Re-scoring pass 1's schedule on the same seed reproduces its score exactly, so a schedule chosen for a lucky score (94.09 on its own seed, about 93.4 held out) is never displaced. Scoring later passes on fresh seeds (mean of 2-3) removes this and makes joint moves of several knots visible: wave 93.50 -> 93.83, then 93.89 with tail rescaling.

F7. A few training steps reveal the conv LR scale, but not the bias LRs (W-2, W-10)

The best LR after 16 steps is 0.1 for conv at every batch size (full-run optimum 0.047-0.22) and within about 10x for head. Bias LRs give flat curves, yet norm_bias matters over a full run (LR 1 instead of 99 costs 0.26 pp) while whiten_bias does not. Freezing LRs that look flat early is therefore harmful (scenario B under ramp: 93.13).

F8. The representation is not the limit under transforms (W-11)

Oracle knots (baseline divided by the transform, grid-snapped) score 93.91 under ramp and 93.97 under wave.

6. What did not work

Idea	Result	Log
Searched cooldown multiplier, 20% lookahead	best 93.78 at batch size 2000	W-5
Prefer a larger LR within x pp of the best	93.54-93.65	W-5
Search the lookahead's level / a geometric shape	93.82 / 93.74 without a transform	W-8
Max of linear and geometric lookahead	93.88 / 93.71 / 93.76	W-11
Full-run knot polish on a single seed	fixes ramp, not wave	W-9
Sloped first segment (first_knot)	wave 93.45-93.61	W-12
Cheap pass 1 (20% lookahead), then pass 2	93.64	W-17
Freeze LRs that look flat in the first scan	scenario B ramp 93.13	W-16

7. Open problems

- Batch sizes 500 and 125 fall short of the new baselines. The gap is in the search, not the representation (F3). At batch size 125 the search's own score exceeds its held-out score by about 0.2 pp, a sign of remaining winner's curse.
- The spread between search seeds is 0.1-0.4 pp, about the size of the differences between the best variants; 2-4 search seeds rank them only coarsely.
- Scenario B costs about 3x scenario A, and cost roughly doubles from batch size 2000 to 125.
- Only multiplicative transforms were tested.

8. Reproduce

Task	Command
Run a search	CIFAR_EXPERIMENTS=auto_a_v1x3m2t_bs2000_ramp CIFAR_SEEDS=0 /venv/main/bin/python train_cifar.py (the log ends with auto_result and heldout_result lines)
Summarize runs	python research/summarize_auto.py [--since YYYYMMDD_HHMMSS] [name ...]
Comparison table	python research/final_table.py
Probes	research/cooldown_fidelity.py (F1, F2), research/overfit_screen.py (F7)
Figures	python research/plot_transforms.py; python research/plot_lookahead_bias.py
This report	python research/make_report.py